

DATA SERIES 16.0

DATA ENGINEERING

`bigquery-public-data.chicago\_taxi\_trips.taxi\_trips`

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Nomor 1

Query:

Untitled query

RUN SAVE DOWNLOAD SHARE SCHE

```
1 SELECT
2   FORMAT_TIMESTAMP('%A', TIMESTAMP(trip_start_timestamp)) AS weekday,
3   AVG(trip_seconds) AS avg_seconds,
4   APPROX_QUANTILES(trip_seconds, 2)[OFFSET(1)] AS median_seconds,
5   STDDEV(trip_seconds) AS stddev_seconds
6 FROM
7   `bigquery-public-data.chicago_taxi_trips.taxi_trips`
8 WHERE
9   EXTRACT(DAYOFWEEK FROM TIMESTAMP(trip_start_timestamp)) != 1
10 GROUP BY
11   weekday
12 ORDER BY
13   median_seconds
14
15
```

Query results

JOB INFORMATION

RESULTS

CHART

JSON

EXECUTION DETAILS

EXECUT

row	weekday	avg_seconds	median_seconds	stddev_seconds
1	Monday	839.4797835193...	540	1346.060208432...
2	Saturday	737.4105515098...	545	1146.335888011...
3	Tuesday	819.6078456596...	549	1302.086733292...
4	Wednesday	841.2479594401...	551	1311.173418881...
5	Thursday	864.1490964545...	588	1312.844235951...
6	Friday	836.5758151211...	588	1257.303982144...

Results:

weekday	avg_seconds	median_seconds	stddev_seconds
Monday	839.4797835	540	1,346.06
Saturday	737.4105515	545	1,146.34
Tuesday	819.6078457	549	1,302.09
Wednesday	841.2479594	551	1,311.17
Thursday	864.1490965	588	1,312.84
Friday	836.5758151	588	1,257.30

Analysis:

If we look at the data above, we could observe the fact that on weekdays, Saturdays not included, Chicago Taxis travel time is longer for each trip compared to in weekend. This implies that on weekdays, passengers are often in need of rides that could possibly of farther distance, or the road is busier compared to the ones in Weekends.

While longer trips also mean longer travel time, I like to believe the second scenario is more likely to happen. Weekdays are the days that are the busiest, since everyone is more likely to go to/off from work. This could make the road more hectic and cause a jam, hence the longer travel time. I would also like to believe that workplaces can often be farther from your living place compared to entertainment areas, like the mall for instance, since when you're looking for a workplace, you might just do anything to get you hired at a place and wouldn't mind the distance it is at as long as it is reachable and not too far away, like a continent away, for most and ideal cases, depends on the job. Longer time travel is usually caused by the long traffic jams at the hours of going to/off work. Since everybody will be on the road at that point in time, the road will be more bustling and more prone to a bottleneck.

Besides that, you could also see that in the median_seconds data column, Monday has the lowest middle point of an average time but also has the higher average_seconds compared to Saturday (a weekend day). This implies most of the trips on Mondays are at the higher end of the bar (travel time) but have a bigger standar deviation value. Meaning, the range between the max and min value of the travel time on Monday is bigger than the one on Saturday with an average of "most" trips ends up on the "longer" side for travel time. Meanwhile, you could see that on Saturday the average travel time is lower than the one on Monday but have more accuracy on it since the standard deviation is lower than the one on Monday. Meaning, the median_seconds value on Saturday is more likely to be closer to the average value compared to the one on Monday.

Speaking of which, you could see the trend is the same as the other weekdays (Tuesday, Wednesday, etc.). The average_seconds is higher of the ones in weekday compared to the one in Saturday. But the standar deviation is also higher on weekdays compared to the one on Saturday. Meaning, even though travel time on weekdays is longer on average, it is inconsistent and fluctuating more compared to the travel time of the one on Saturday. Meanwhile, the travel time on Saturday is more consistent, consistently shorter on travel time compared to the one on weekdays.

Nomor 2

Query:

```
1 SELECT
2   pickup_community_area,
3   dropoff_community_area,
4   COUNT(DISTINCT(taxi_id)) AS num_trips
5 FROM
6   `bigquery-public-data.chicago_taxi_trips.taxi_trips`
7 WHERE
8   pickup_community_area IS NOT NULL AND
9   dropoff_community_area IS NOT NULL AND
10  EXTRACT(YEAR FROM trip_start_timestamp) = 2023 AND
11  TIMESTAMP_DIFF(trip_end_timestamp, trip_start_timestamp, MINUTE) > 10
12 GROUP BY
13   pickup_community_area,
14   dropoff_community_area
15 ORDER BY
16   num_trips DESC
17 LIMIT 5
18
```

Query results

JOB INFORMATION		RESULTS	CHART	JSON	EXEC
Row	pickup_community_area	dropoff_community_area	num_trips		
1	8	8	2935		
2	32	8	2915		
3	8	32	2912		
4	8	28	2884		
5	32	32	2866		

Results:

pickup_community_area	dropoff_community_area	num_trips
8	8	2935
32	8	2915
8	32	2912
8	28	2884
32	32	2866

Analysis:

From the data above here, we could see that the “go to” spots are the community area: 8, 28, and 32, since there’s numerous ammount of trips that often goes there in 2023. We could assume from that information alone, those community areas are either a really popular area or a workplace of many, or even both. There isn’t really anything much that you can get from just that data alone, but we could just assume that community areas 8 and 32 are most likely connected in some sort of way, since they’re the ones with the greatest number of trips and the ones that have shown up the most in the top 5 data of routes with the most amount of trips.

Nomor 3

Query:

Untitled query

RUNSAVEDOWNLOADSHARESCHEDULE

```
1 SELECT
2   payment_type,
3   AVG(fare) AS average_fare,
4   AVG(tips) AS average_tips,
5   AVG(tolls) AS average_tolls
6 FROM
7   `bigquery-public-data.chicago_taxi_trips.taxi_trips`
8 WHERE
9   EXTRACT(YEAR FROM trip_start_timestamp) = 2019
10 GROUP BY
11   payment_type
12 ORDER BY
13   average_fare DESC
14 LIMIT 9
15
```

Query results

JOB INFORMATIONRESULTSCHARTJSONEXECUTION DETAILSEXECUT

Row	payment_type	average_fare	average_tips	average_tolls
1	Prepaid	19.46141479099...	0.0	0.0
2	Credit Card	16.81413780718...	3.7745446963091	0.002266943412...
3	Prcard	16.13000271387...	0.203527591108...	0.001529141897...
4	Mobile	15.97317403733...	3.112698023253...	4.968012969128...
5	Unknown	15.87237993336...	0.082129280999...	0.000147177000...
6	No Charge	15.75350963899...	0.249531863948...	0.019141699410...
7	Dispute	15.65584022445...	0.001445658594...	0.078991435321...
8	Cash	12.97830635161...	0.002387440611...	0.001790926027...
9	Pcard	11.32747148288...	0.038022813688...	0.0

Results:

payment_type	average_fare	average_tips	average_tolls
Prepaid	19.46141479	0	0
Credit Card	16.81413781	3.774544696	0.002266943412
Prcard	16.13000271	0.2035275911	0.001529141897
Mobile	15.97317404	3.112698023	0.00004968012969
Unknown	15.87237993	0.082129281	0.0001471770007
No Charge	15.75350964	0.2495318639	0.01914169941
Dispute	15.65584022	0.001445658594	0.07899143532
Cash	12.97830635	0.002387440611	0.001790926027
Pcard	11.32747148	0.03802281369	0

Analysis

From the data above we could see a lot of information regarding what method of payment passengers use and how much they need to pay from using that method of payment. We could see that the prepaid one is the most expensive one for fares, while it also means that they do not need to pay for tolls, nor do they tip. Meanwhile, credit card, pcard, and mobile can be grouped into the second group that has the biggest fares, with credit card and mobile method of payment are the two method of payment that tip the biggest out of all method of payment. Other than that, there is also the unknown and no charge method of payment which can be grouped into the third group that has the biggest fares, with very similar numbers on tippings and tolls. And then there's the last group method of payment, which include cash and pcard method of payment, where they pay the lowest fares, and even have the two lowest number of tips and tolls paid.

From this method of payment, we could also see the class of people that often ride Chicago Taxi and what kind of trips do they usually take, long ones or short ones. We could see that prepaid could belong to the higher end of the economy class or to those who need to travel farther than most, since their fares are higher on average. And for the second and third group, it can be said that they are the middle high and low class, where they just use taxis for their needs. And meanwhile the last group could be said to be the method of payment where you could save money the most and might also mean that they don't really need to travel that far since their fares are also smaller than most.